HIGHER DIRECT IMAGES OF LQC-FBC MODULES COMPUTED ON BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file stacks-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites the lemmas recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Base change of higher direct images. Sheaves on Stacks, Lemma stacks-sheaves-lemma-base-change-higher-direct-images. First reduction.
- Comparison for representable morphisms. Sheaves on Stacks, Lemma stacks-sheaves-lemma-compare-morphism-cohomology. Scheme-case identity.
- Etale computation of R^i f_*. Cohomology of Algebraic Stacks, Proposition stacks-cohomology-proposition-loc-qcoh-flat-base-change. Etale vs fppf computation.

Source reference. Original reference 075H: sources/stacks-chow/tags.txt, source lines 9131–9131. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original proposition 0771: sources/stacks-chow/tags.txt, source lines 9186–9186. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 075N: sources/stacks-chow/tags.txt, source lines 9137–9137. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$ be a morphism of algebraic stacks which is quasi-compact, quasi-separated, and representable by algebraic spaces. Let $\mathcal{F}$ be in $LQCoh^{fbc}(\mathcal{O}_\mathcal{X})$. Then for an object $y : V \to \mathcal{Y}$ of $\mathcal{Y}$ we have*

$$(R^i f_* \mathcal{F})|_{V_{étale}} = R^i f'_{small,*}(\mathcal{F}|_{U_{étale}})$$

where $f' : U = V \times_\mathcal{Y} \mathcal{X} \to V$ is the base change of f.

Proof. By Sheaves on Stacks, Lemma 075H we can reduce to the case where $\mathcal{X}$ is represented by U and $\mathcal{Y}$ is represented by V. Of course this also uses that the pullback of $\mathcal{F}$ to U is in $LQCoh^{fbc}(\mathcal{O}_U)$ by Proposition 0771. Then the result follows from Sheaves on Stacks, Lemma 075N and the fact that $R^i f_*$ may be computed in the étale topology by Proposition 0771. □

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